Outlook

Re: RHD modelling work meet Thursday Oct 1st

From Neema Minja <nminja@uw.edu>
Date Tue 11/3/2026 10:45 AM
To William R Garcia <wrgg@uw.edu>; David Watkins <davidaw@uw.edu>

1 attachment (745 KB)
Liesl et al RHD progression.pdf;

Hi William,

Thanks. Regression in over 20 years is possible but sparsely documented - studies available are few and weak. See attached from Liesl some 10 years back. Small numbers, they thought those with RHD and over 20 were more likely to regress (discussion). Cannon paper has tables at the very end that split rates to very young and 15 - 24 year group (not very informative for true adults). I see some recent papers grouping stage A/B together in reporting as mild RHD (Roselyns GOAL follow up paper).

A follow up question - for the rates, does it matter if these adults started in the <20 yr camp or not? Liesl paper above started with teens and went back 5-6 years to check status, by which time they were adults. I think most studies are like that - we don't really have true adult. Just to highlight the 20 cut-off is for diagnosis (the reason behind is that borderline presents a sprectrum of changes that can be normal and specificity of the criteria is greater when stricter rules are applied to adults). I wonder if it has been thought about in terms of regression etc.
David may be right that for older adults diagnosed after 20, regression probability is small. But there is some evidence regression is happening in some form in those diagnosed as adolescents and followed up to >20 years for mild disease.

I need to put a 30 mins placeholder for this Thursday - what time is good? Will send soon.

Best,
Neema

From: William R Garcia <wrgg@uw.edu>
Sent: 02 November 2026 16:56
To: Neema Minja <nminja@uw.edu>; David Watkins <davidaw@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Hi Neema,

I had a quick review discussion with David. Based on that could you help us clarify:

For adults aged 20+, should we allow regression from stage B directly to no RHD, given that stage A does not apply? Or should we assume no regression from B in adults?

David suggested provisionally setting this regression rate to zero, pending your advice. If you think regression should be allowed, is there evidence or an approximate annual rate you would recommend?

Best,

William

From: Neema Minja <nminja@uw.edu>
Sent: Sunday, November 1, 2026 10:38 PM
To: William R Garcia <wrgg@uw.edu>; David Watkins <davidaw@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Thank you, William.

Glad you are feeling better. I will review this and revert.

Thanks,
Neema

From: William R Garcia <wrgg@uw.edu>
Sent: 31 October 2026 14:15
To: Neema Minja <nminja@uw.edu>; David Watkins <davidaw@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Dear Neema and David,

I wasn't feeling well yesterday, but I'm better today and had the chance to create the mock models. Attached is the entry-age version for discussion before updating the codebase.

The model separates childhood- and adult-onset pathways, retains those pathways as people age, and incorporates changes in screening and BPG adherence over time. Parameters remain illustrative and require calibration.

Please review these priorities (in particular yellow cells in Excel):

- **Stage B regression:** We currently assume 30% regression over two years for childhood-onset Stage B, routed to A, and extrapolate a lower rate for adult-onset B, routed directly to No RHD. These assumptions produce much faster regression than the Cannon-derived alternative.
- **D→C:** We currently assume D→C = 0 in the codebase. I think is OK, WHF stage D represents clinical complications, not simply a worse echocardiogram than C, so if want to change introducing a fixed "valve regression" rate requires justification.
- **Screening and SAP:** Please review the screening-target interpretation and the current assumption that residual Stage A cases stop BPG after age 20.

Please give it a look and share your feedback. I'll update the codebase and run the model accordingly.

Best,

William Garcia
PhD Candidate, Health Metrics & Implementation Science
University of Washington

From: William R Garcia <wrgg@uw.edu>
Sent: Friday, October 30, 2026 10:42 AM
To: Neema Minja <nminja@uw.edu>; David Watkins <davidaw@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Thanks Neema,

This is super claryfing.

Best,

William

From: Neema Minja <nminja@uw.edu>
Sent: Friday, October 30, 2026 8:31 AM
To: William R Garcia <wrgg@uw.edu>; David Watkins <davidaw@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Just remembered one other thing on the effect of BPG - incorporating the cascade of care (which can be country individualized) such that with good BPG adherence, progression is much slower. (GOAL's 0.8% vs. 8% progression with perfect BPG adherence vs. none)

The main summary today was to use a systematic approach -
Use two pathways - those who enter with age =<20 years and those who enter at > 20 years .

Roughly;
For those =<20 years,
Progression is from No RHD A-B-C-D, regression C-B; B-A; A-no RDH
Composition Stage A/B/C/D - 50%/30%/15%/5%
Progression increases as you move up stages, A to B < B to C, < C to D
And regression does the opposite, decreases as you move up stages
A to No RHD > A to B > B to C, > C to D
(We can use the WHF for stage A and B regression (50% and 30%) I think need to be annualized it was a 2-year period. If unsure about specific rates, we can dig up from my Excel.

Once someone starts the younger cohort, they continue to be subject to TPs relevant to that cohort. The GOAL POST paper showed children from GOAL followed up to 25 years some still regressed to no RHD.

For those > 20 years,
Progression is from No RHD -B-C-D, regression C-B; B-A; A - No RHD
Composition Stage B/C/D - 60/30%/10%
Same pattern of progression and regression, starting from stage B.
Rates different from those =<20 years (I would use Claude to ask specific questions on extrapolating data from Cannon paper)

For screening, frequency of screening and age- discussion with David later today to operationalize how to translate targets to annual screening rates. (Focus on 0 - 39 years between now and 2050) and cumulative effect on impact and costs

Thanks so much,
Neema.

From: Neema Minja <nminja@uw.edu>
Sent: 29 October 2026 21:10
To: William R Garcia <wrgg@uw.edu>; David Watkins <davidaw@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

William,

Thanks, looking forward to the chat.

Neema.

From: William R Garcia <wrgg@uw.edu>
Sent: 28 October 2026 17:40
To: David Watkins <davidaw@uw.edu>; Neema Minja <nminja@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Hi Neema and David,

With help from Claude, I went through the expert meeting notes and added comments to the Word document (attached) based on current model state. I hope you can look at them before we talk on Friday.

For the poster, I suggest we work in clarifications for improving two blocks:

1. Policy implementation details: the design of the screening campaign and the SAP rules by stage, including age limits and whether stage D patients continue treatment.
2. Calibration: recalibrating the model with the revised mortality and regression estimates, taking their follow-up periods into account.

Looking forward to discussing.

Best,

William

From: David Watkins <davidaw@uw.edu>
Sent: Wednesday, October 28, 2026 3:39 PM
To: William R Garcia <wrgg@uw.edu>; Neema Minja <nminja@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Great - Neema please send an invite for 7am then

From: William R Garcia <wrgg@uw.edu>
Date: Wednesday, October 28, 2026 at 3:34 PM
To: Neema Minja <nminja@uw.edu>; David Watkins <davidaw@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Hi Neema and David:

I am also available; please let's discuss this, and I will then make the necessary adjustments and re-run the process for the poster presentation.

I suggest 7 AM Seattle time if possible

Best,

William

From: Neema Minja <nminja@uw.edu>
Sent: Wednesday, October 28, 2026 3:25 PM
To: David Watkins <davidaw@uw.edu>; William R Garcia <wrgg@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Thank you,

A meeting will make some good progress- I will be available within those time slots on Friday.

NM

From: David Watkins <davidaw@uw.edu>
Sent: 28 October 2026 15:06
To: Neema Minja <nminja@uw.edu>; William R Garcia <wrgg@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Thanks, Neema. Let's have a quick meeting to make a game plan on your points below and re-run once more before poster.

Could you and William do Friday sometime in the 6-8am Seattle time slot?

DW

From: Neema Minja <nminja@uw.edu>
Date: Tuesday, October 27, 2026 at 2:07 PM
To: William R Garcia <wrgg@uw.edu>; David Watkins <davidaw@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Dear William and David,

I had ta great discussion with the experts on Thursday and it was very informative! Time was not enough actually but we tackled some of the pressing questions. I will follow up with them over email on one or two of the questions and any other things unclear moving forward.

The issue of screening - in favour of screening once between ages 5 - 19 years. Discussed with David on an idea he had of including costs of screening whole population at the beginning ages 5 - 39 years and then after that you screen once ages 5 - 19 years.

How long on SAP - well, data shows mild disease you don't have to continue for long, those who normalize do not go on to develop severe disease, so I was wondering about sub-setting the length of the prophylaxis etc. For other groups, BPG is personalized, often long term for severe lesions(I assume until 40 years/ stage C), except when irreversible with complications.
Mortality - much higher. Yes, they all agreed logical sequence and mortality is from Stage D/complications, even if it's just for 5 minutes etc. The main point was mortality once you are in stage D in LMIC is very high! Like 25% per year minimum (e.g. REMEDY sub-analysis in Namibia)

One of the points that came out was to separate TPs from literature that reported good adherence from those that had suboptimal or didn't report adherence. So, there is a possibility of separating TPs based on adherence rates. They commented on regression rates for mild and moderate disease, which had good evidence to back up in literature (see document).
As a side note - there is a big difference between some of the Indonesia/Uganda TP rates.

What are your thoughts? Could we do another run with some of these effected. The two big ones are the screening and SAP?

I am working out the visa and other travel plans but have to finalize the presentation poster next week.

From: Neema Minja <nminja@uw.edu>
Sent: 22 October 2026 23:49
To: William R Garcia <wrgg@uw.edu>; David Watkins <davidaw@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Dear William,

Thank you, I see what you mean. GBD mortality <20 is already thought to be too low so we cannot have lower than that in the model. Tomorrow we will focus on parameters; I think results might change once we have some expert input and agree on assumptions. The three of us could also plan to meet after this meeting within the next two weeks as well.

Thanks,
Neema

From: William R Garcia <wrgg@uw.edu>
Sent: 22 October 2026 17:27
To: Neema Minja <nminja@uw.edu>; David Watkins <davidaw@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Hi Neema and David,

Please find attached the updated slides, incorporating the revised screening populations and age-specific frequencies.

I tried adjusting the calibration, but childhood mortality is still clearly off—see slide B9. Given the age restriction on stage A and the limited adult evidence, might it be better to distinguish the states,

progression, and mortality assumptions for children/younger and older age groups? Let's wait for tomorrow's discussion before making further changes. Could you also discuss incorporating recommended SAP durations at a later stage?

In case I won't be able to attend tomorrow's early meeting, as I expect to be working late tonight. Could you please save the meeting transcript, if possible, so I can catch up afterward?

Thanks,

William

From: Neema Minja <nminja@uw.edu>
Sent: Thursday, October 22, 2026 8:21 AM
To: William R Garcia <wrgg@uw.edu>; David Watkins <davidaw@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Thanks David and William,

William - thanks for the slides. My questions can wait for next week.
Let's see what results from adjusting the screening frequencies and ages. I will update the slides with those results and share for tomorrow's meeting. For SAP coverage, apart from constraining withing screening coverage, would we incorporate recommended durations at a later stage?

Looking forward to the chat tomorrow. William, feel free to attend or not to. We will just focus on the parameters.

Thanks,
Neema.

From: William R Garcia <wrgg@uw.edu>
Sent: 21 October 2026 21:28
To: David Watkins <davidaw@uw.edu>; Neema Minja <nminja@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

This makes sense, with this adjustment the BCR will improve substantially since the costs will be better captured.

I will run the updates tomorrow morning

Thanks David

Get Outlook for Android

From: David Watkins <davidaw@uw.edu>
Sent: Wednesday, 21 October 2026 21:15:39
To: William R Garcia <wrgg@uw.edu>; Neema Minja <nminja@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Looks great.

Glad calibration is working better - I'd still like to see the details when we meet.

The SAP population treated/costed is straightforward: RHD prevalence x coverage %. This is specified according to a cascade of care a la HTN cascade, at a later stage in the project. For now just prevalence x coverage is good enough.

The screening population is something to discuss. For now, please adjust as follows:

1. Assume that children are screened twice between ages 5 and 19, so the expected number of screenings in a given year is 2/15 = 0.1333. So multiply 5-19 population size by 0.1333 (treated proportion) and by coverage % to get the cost in a given year
2. Assume adults are screened once between age 20 and 39, so the expected number of screenings in a given year is 1/20 = 0.04. So multiply 20-39 population size by 0.04 and by coverage %
3. The sum of child and adult screening cost is total screening cost
4. SAP coverage % is constrained within screening coverage % (obviously)

Maybe try re-running things and sending another version of the slide deck Thursday? I suspect the BCRs will be much more favorable due to lower cost.

Neema has her own slide deck for her expert meeting Friday and can include some of the updated slides as prelim results.

DW

From: William R Garcia <wrgg@uw.edu>
Date: Wednesday, October 21, 2026 at 8:49 PM
To: David Watkins <davidaw@uw.edu>; Neema Minja <nminja@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Hi Neema and David,

Please find attached the current technical slides. Calibration is working better, particularly for projected deaths—please see slide 27 for the updated trends.

The eligible population still needs attention for a credible investment case. WHO recommends echo screening among ages 5–19 in populations with moderate/high RHD prevalence, rather than annual whole-population screening. SAP recommendations also differ by age and diagnosis pathway, while WHF distinguishes individualized prophylaxis for stage A from recommended prophylaxis for stage B in those aged ≤20.

How would you suggest parameterizing screening ages and frequency, SAP eligibility, and treatment duration? Would separate pathways for screen-detected cases and clinically diagnosed patients make sensey?

This is still work in progress. Please feel free to look through the full deck and flag anything else that seems off. The VSL + GDP benefit formula is incorrect, and I am already fixing it.

Best,

William

From: David Watkins <davidaw@uw.edu>
Sent: Friday, October 16, 2026 2:21 PM
To: Neema Minja <nminja@uw.edu>; William R Garcia <wrgg@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Great start. I added a few comments to your slides, please have a look. I chatted with William, who will incorporate your edits to the model/input data early next week before the meeting, and you can then update the slide deck. Has the expert meeting day/time been confirmed?

DW

From: Neema Minja <nminja@uw.edu>
Date: Thursday, October 15, 2026 at 3:39 PM
To: David Watkins <davidaw@uw.edu>; William R Garcia <wrgg@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Dear David and William,

Sharing a slide deck for the discussion
RHD SAP investment tool 1.pptx
Fyi – The transition rates come from studies with various follow up times, on slide 7, I have put annualized ranges in column 3. I just don't know if the ranges in the first table will be helpful (probably less useful than the figures in the next few slides that might work better).

Factors such as SAP use (yes/no); optimal vs. suboptimal etc. affect progression. A good example from GOAL study control vs. Trial arm. So that definitely impacts the rates and explains partly the heterogeneous results. For the model, William did we talk about this? TP rates changing with improved adherence? and did you want anything else to be discussed? Anything on the table with calibrated A/B/C/D stage structure, implied case fatality, and the excess-death attribution?

Finally, should we add result figures (from output? slide 12 placeholder) - e.g. Projected impact: reference vs SAP scale-up – incidence, prevalence and deaths or the annual + cumulative RHD deaths averted etc.

Thanks,
Neema

From: Neema Minja <nminja@uw.edu>
Sent: 30 September 2026 21:42
To: David Watkins <davidaw@uw.edu>; William R Garcia <wrgg@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

Interesting - thanks. Chat soon.

From: David Watkins <davidaw@uw.edu>
Sent: 30 September 2026 21:37
To: Neema Minja <nminja@uw.edu>; William R Garcia <wrgg@uw.edu>
Subject: Re: RHD modelling work meet Thursday Oct 1st

I think that's right - they are also doing some modeling on value of information relating to R&D for a group A strep vaccine. Less emphasis on RHD, from what Andrea told me. But good to compare notes and align efforts where relevant.

From: Neema Minja <nminja@uw.edu>
Date: Wednesday, September 30, 2026 at 8:57 PM
To: David Watkins <davidaw@uw.edu>; William R Garcia <wrgg@uw.edu>
Subject: RHD modelling work meet Thursday Oct 1st

Hi David and William,

I wanted to check in on tomorrow's meeting with Pete and Tom. It will be at 8am PT. It looks like Pete is an experienced professor in mathematical modelling - working on a vaccine model in the same area ARF/RHD (we will hear more tomorrow)

My understanding is the main agenda is to exchange ideas on both ends on the models we are working on and any other specific aspects.

Chat soon,
Neema